

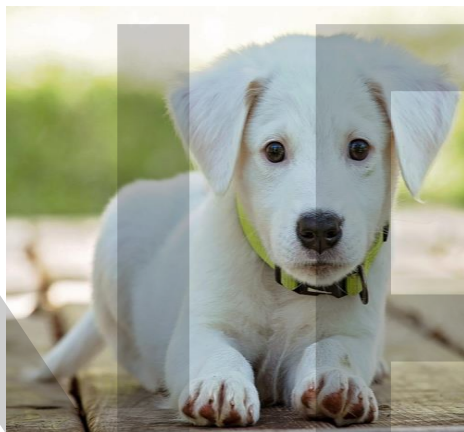
Object Detection

Contents

- ✓ Introduction to Object detection
- ✓ Selective Search Algorithm
- ✓ Region proposals
- ✓ R-CNN framework
- ✓ Fast R-CNN framework
- ✓ Faster R-CNN framework

Classification, Localization and Object Detection

Classification



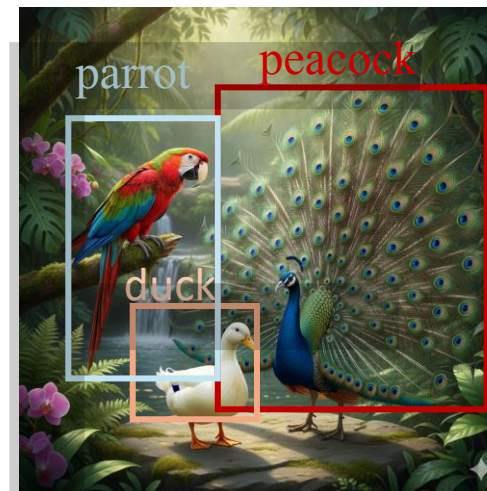
One single class label for the whole image (dog class)

Localization



Box in the image(x, y, w, h)

Object Detection



Object Detection = Classification + Localization

x, y: Coordinates of the centre of the bounding box
h,w: height and width of the image

Localization as a Regression Problem



Neural network

Bounding box coordinates(x, y, w, h)
(predicted values, 4 numbers)

Ground Truth

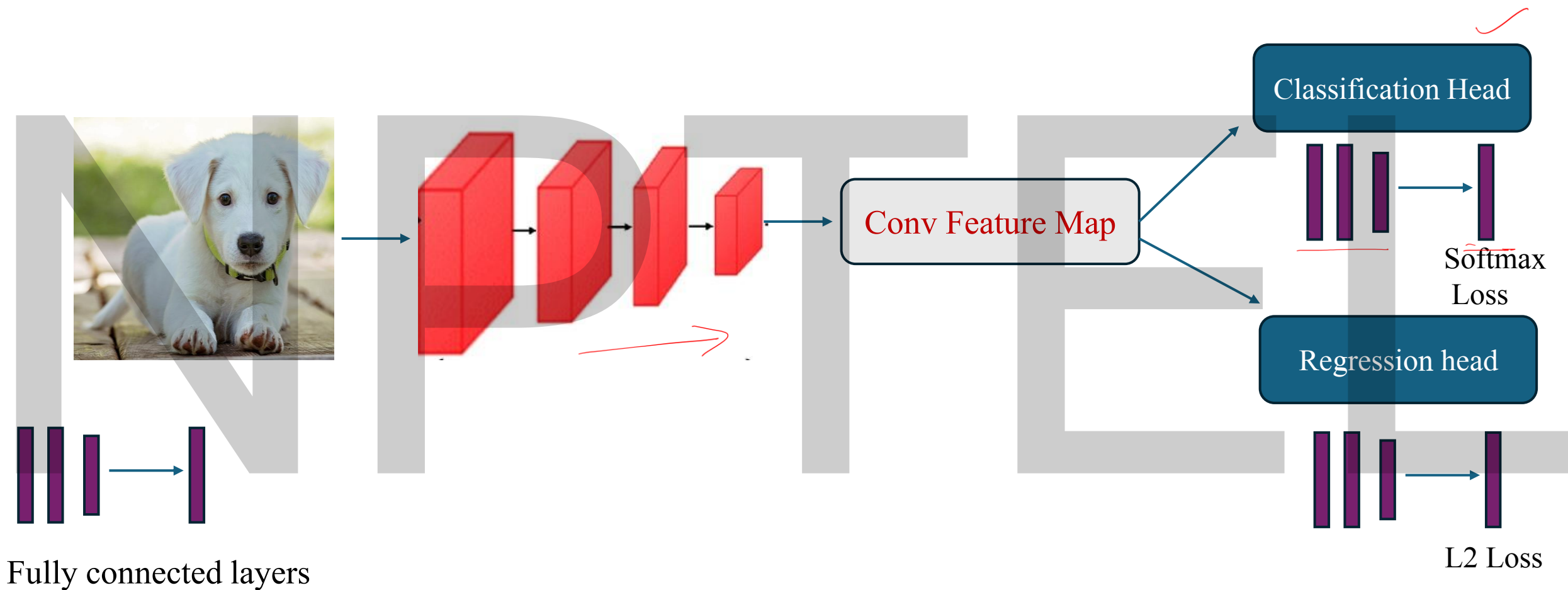
Correct bounding box coordinates

Error

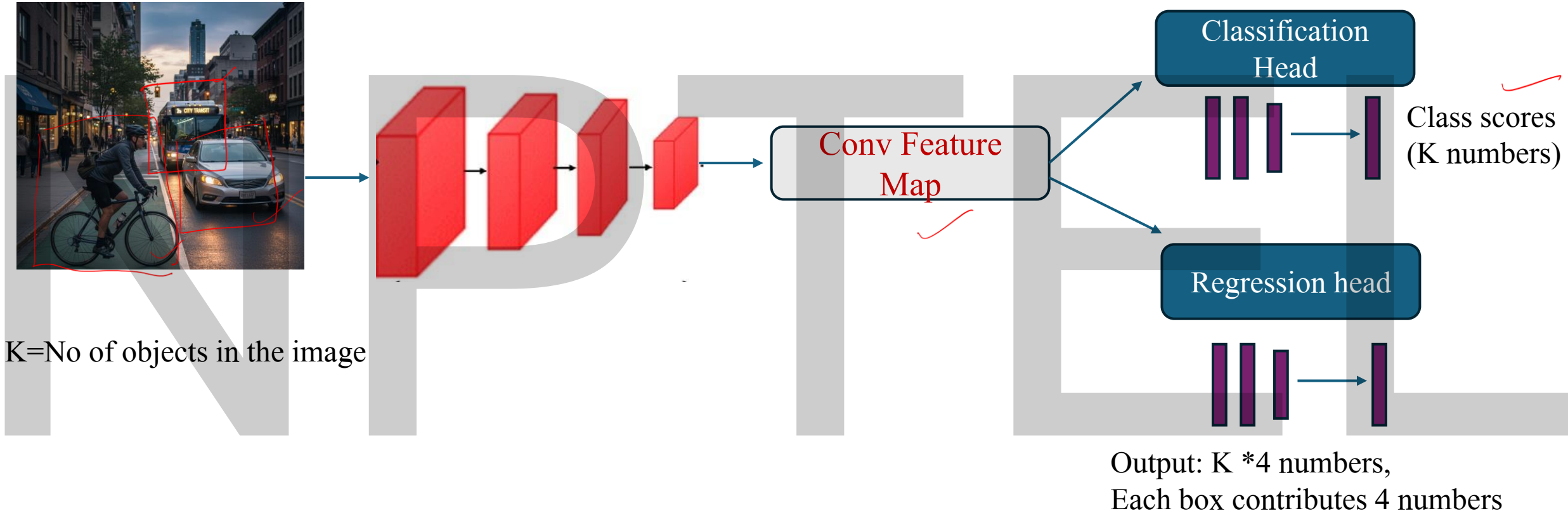
L2 distance between the predicted values and the ground truth

$$\sqrt{\cancel{x_2} - \cancel{x_1}^2 + \cancel{y_2} - \cancel{y_1}^2}$$

Can Classification and Localization be done together?



Multiple Object Detection



Can CNNs trained for classification be used for object detection?

Using Sliding Window

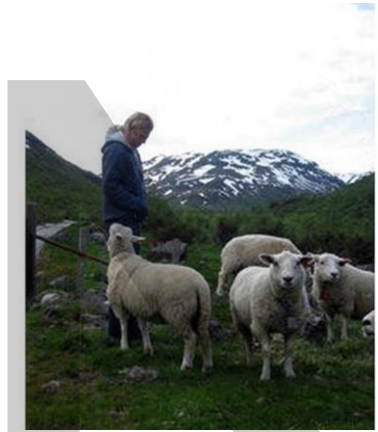


Cat ? Cat ? Cat ? Cat ? Cat ?
Dog ? Dog ? Dog ? Dog ? Dog ?

Multiple positions to be scanned which is computationally expensive

Selective Search Algorithm

Begins with over-segmentation of the image called super pixels

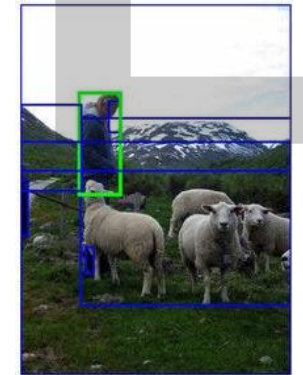
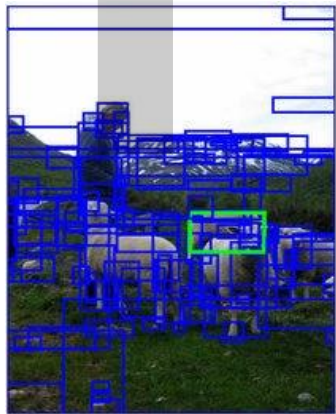
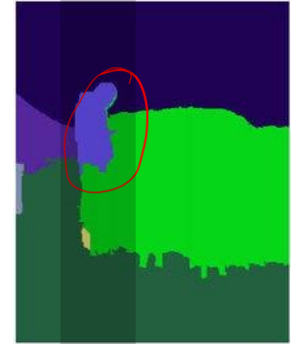


The bounding box of this region is recorded as a **region proposal**.



Merges **similar neighboring** regions repeatedly creating a larger region.

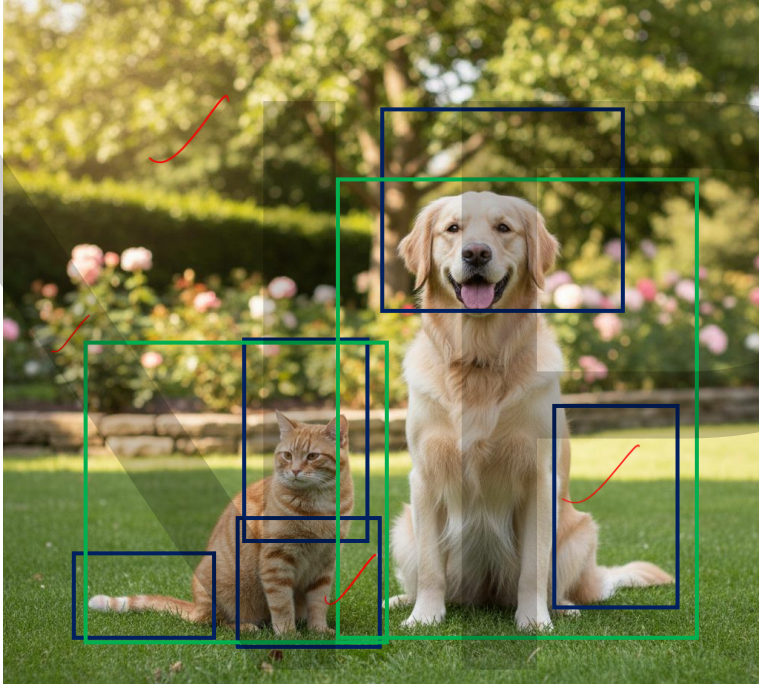
- ✓ Color
- ✓ Texture.
- ✓ Size similarity
- ✓ Shape Compatibility



Ref:

Uijlings, J.R., van de Sande, K.E.A., Gevers, T. and Smeulders, A.W.M. (2013). Selective Search for Object Recognition. *International Journal of Computer Vision*, **104**(2), 154-171.

Region Proposals- Based on Selective Search Algorithm

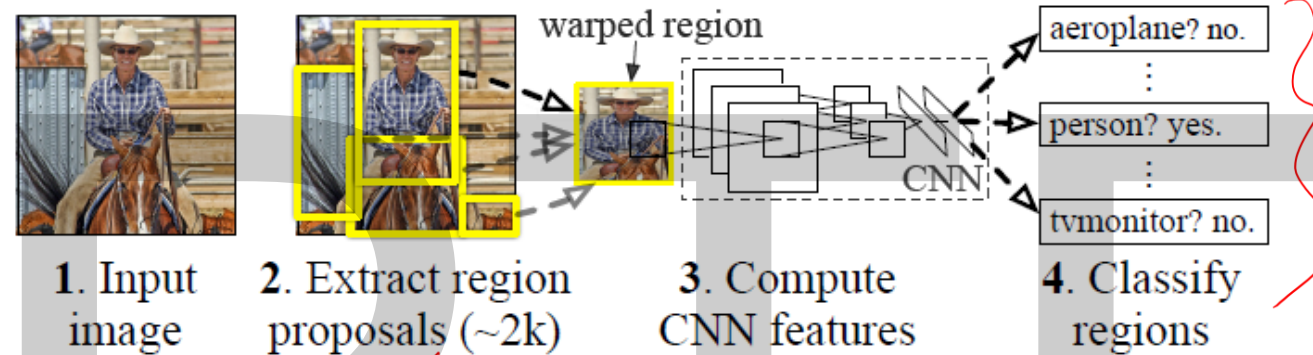


- ✓ Find regions that are likely to contain an object
- ✓ Class agnostic object detection

Blue boxes are false positive and green boxes are true positive

R-CNN Framework

R-CNN: *Regions with CNN features*



- ✓ First proposes candidate regions in an image.
- ✓ Feed these regions through a convolutional neural network (CNN) to extract rich features.
- ✓ Classifying each region using class specific linear SVMs.

Ref:

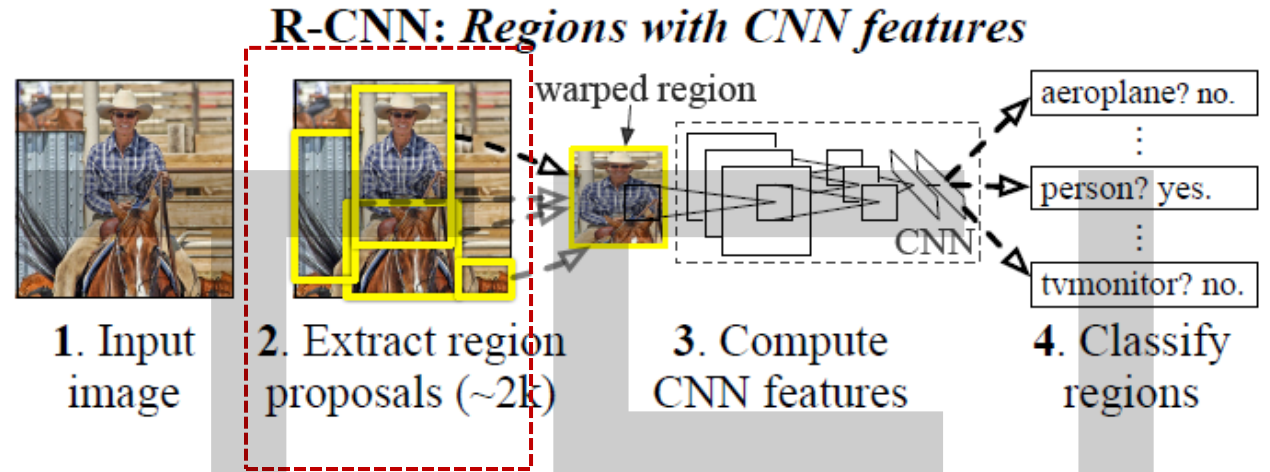
Rich feature hierarchies for accurate object detection and semantic segmentation

Ross Girshick, Jeff Donahue, Trevor Darrell, Jitendra Malik (2014) [arXiv:1311.2524v5](https://arxiv.org/abs/1311.2524v5) [cs.CV]

R-CNN-Training

Region Proposal Generation

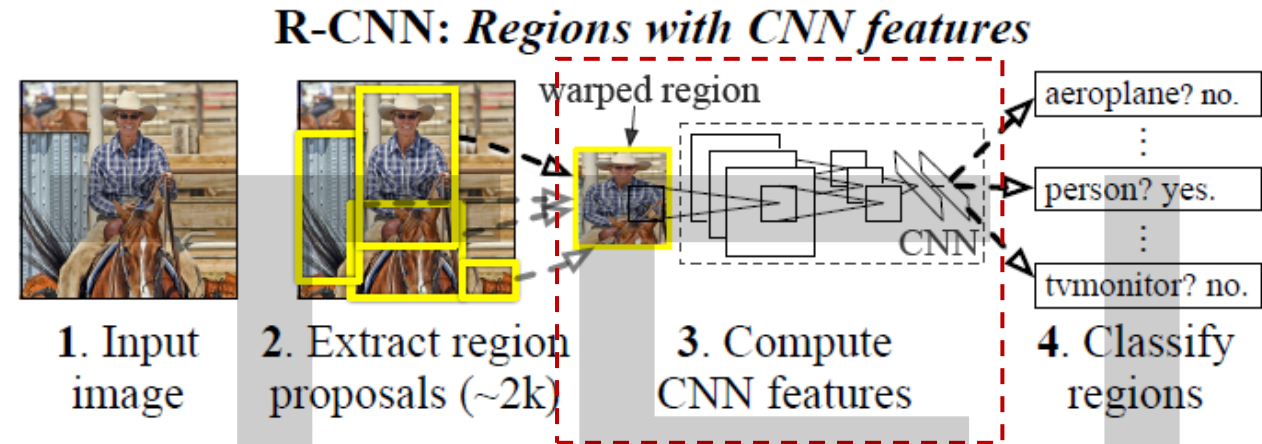
- ✓ Category-independent Selective Search.
- ✓ Provides candidate bounding boxes for objects.
- ✓ Produce a set of candidate regions such that almost every true object in the image overlaps reasonably well with at least one candidate box.



R-CNN Training

Feature Extraction with CNN

- ✓ Download classification model Alexnet.
- ✓ Finetune it for Pascal VOC 2007 dataset which has 20 object classes.
- ✓ Crop each region from the original image.
- ✓ Warp/resize each crop to a fixed size (227×227 for AlexNet).
- ✓ Pass each region through a CNN to extract a feature vector (4096-D features)



Feature Extraction from Region Proposals

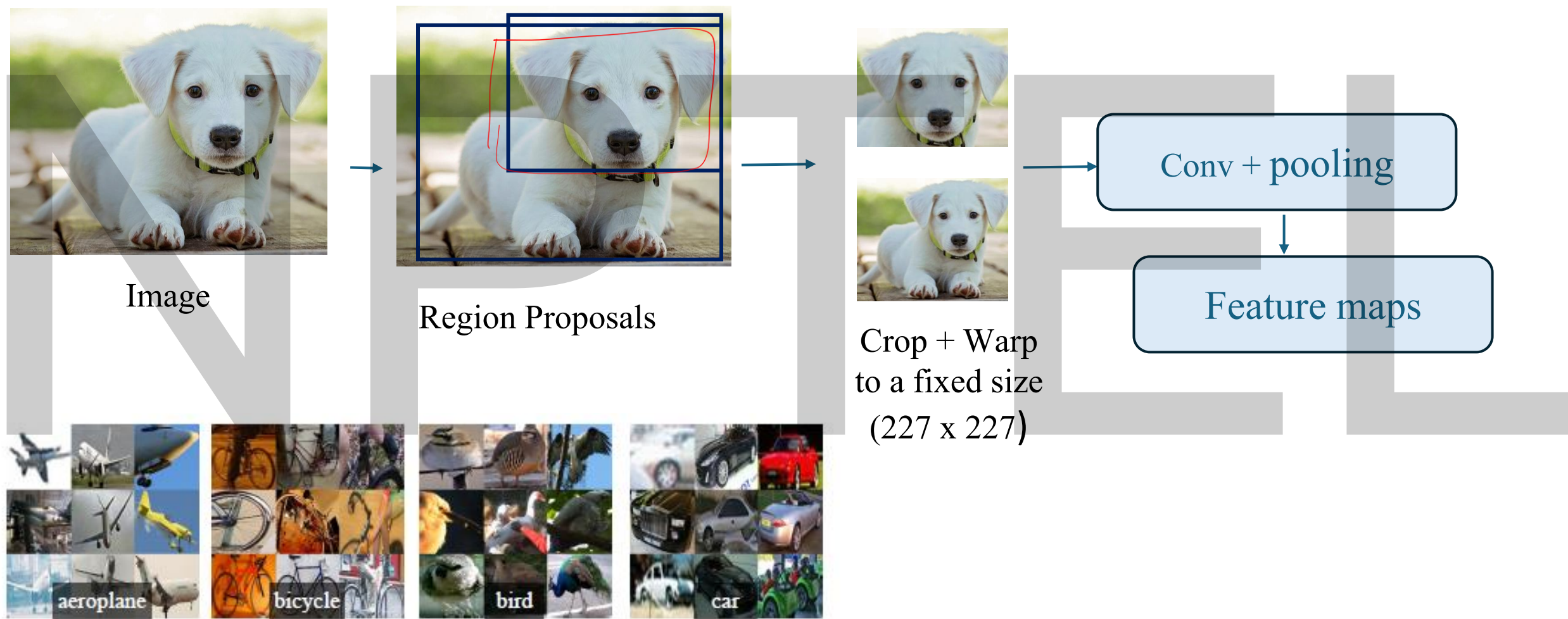


Figure 2: Warped training samples from VOC 2007 train.

R-CNN Training

Class-Specific SVMs

- ✓ Train one **linear SVM per class** using extracted features.

Binary SVM for Each class



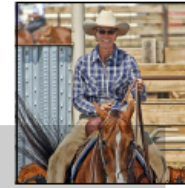
Positive samples of Flower SVM

Positive samples of dog SVM

Positive sample of
Cat SVM

Negative samples of Cat SVM

R-CNN: *Regions with CNN features*

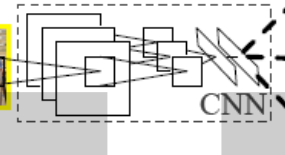


1. Input image



2. Extract region proposals (~2k)

warped region

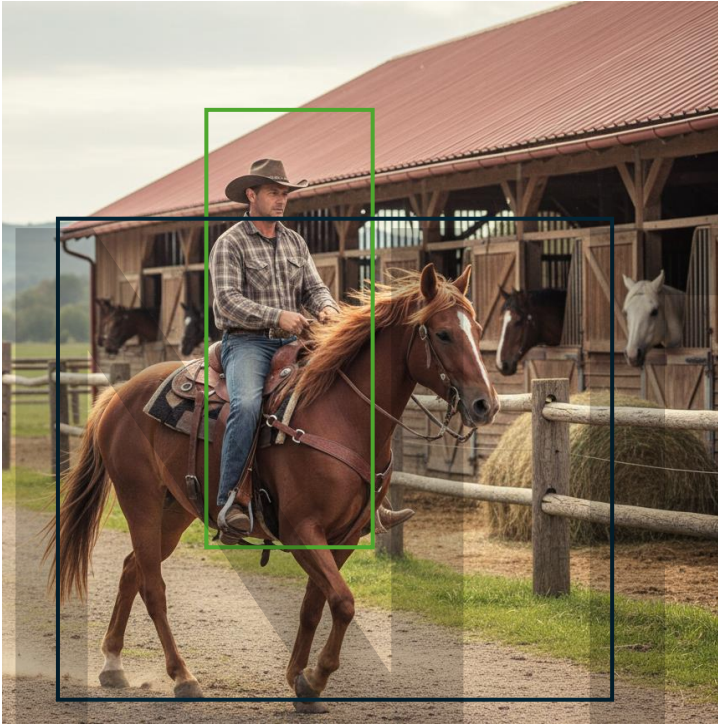


3. Compute CNN features

aeroplane? no.
person? yes.
tvmonitor? no.

4. Classify regions

Training with SVMs



Two classes of objects:

- ✓ Human Figure
- ✓ Horse



There will be two SVMs:

- Horse Class
- Human Class



- ✓ Feed the features extracted from CNN to **ALL SVMs**.
- ✓ Each SVM outputs a score.
- ✓ The class with highest positive score wins
- ✓ If all SVM scores are negative → background

Bounding Box Regression



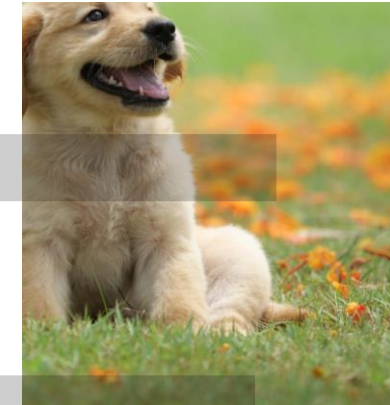
Correct Proposal



Proposal too much left



Proposal too much wide

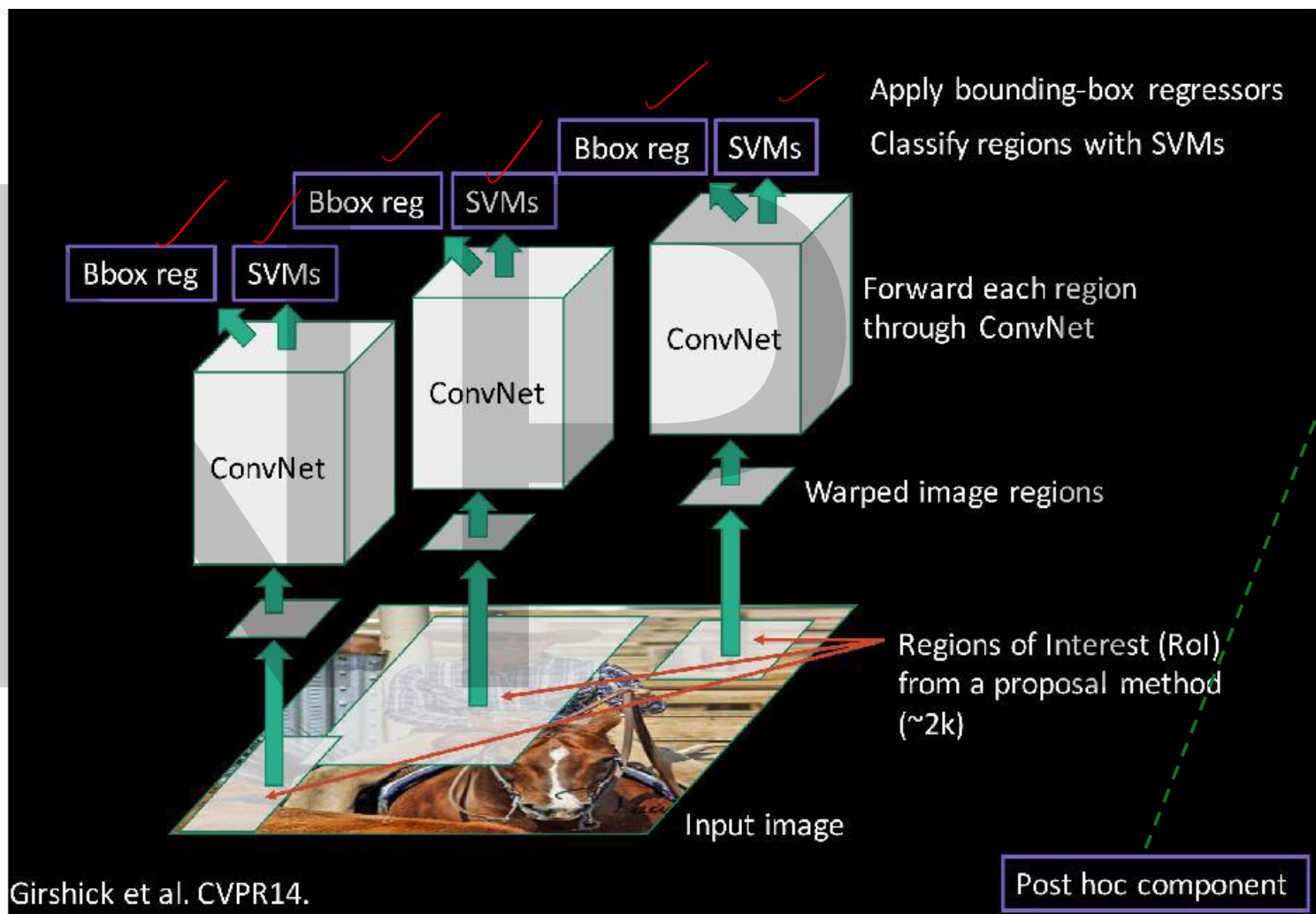


Proposal without the full dog's image

Regressor predicts the *offsets* (dx , dy , dw , dh) between the proposed box and the ground truth box.
This is used to compute the refined box

The bounding box regression helps improve the accuracy of object localization by adjusting the initially proposed bounding box to better fit the object's actual boundaries.

R-CNN Framework



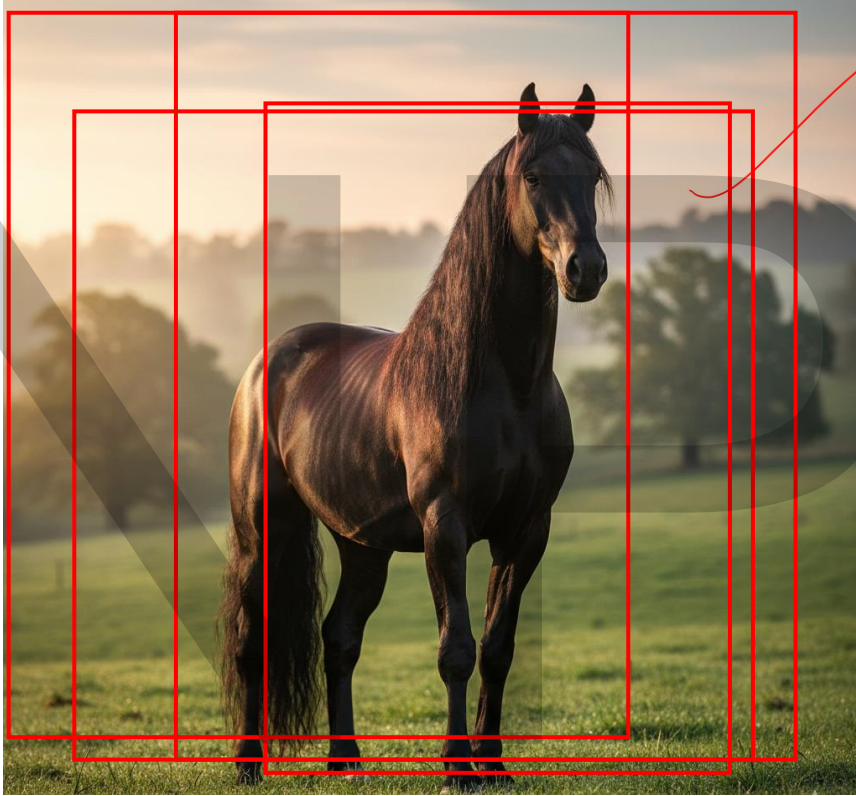
After extracting features for each RoI, **SVMs** are trained separately for each class.

A separate **linear regressor** is trained independently to refine the bounding box coordinates.

Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014

Slide credit: Ross Girshick

Non-Maximum Suppression



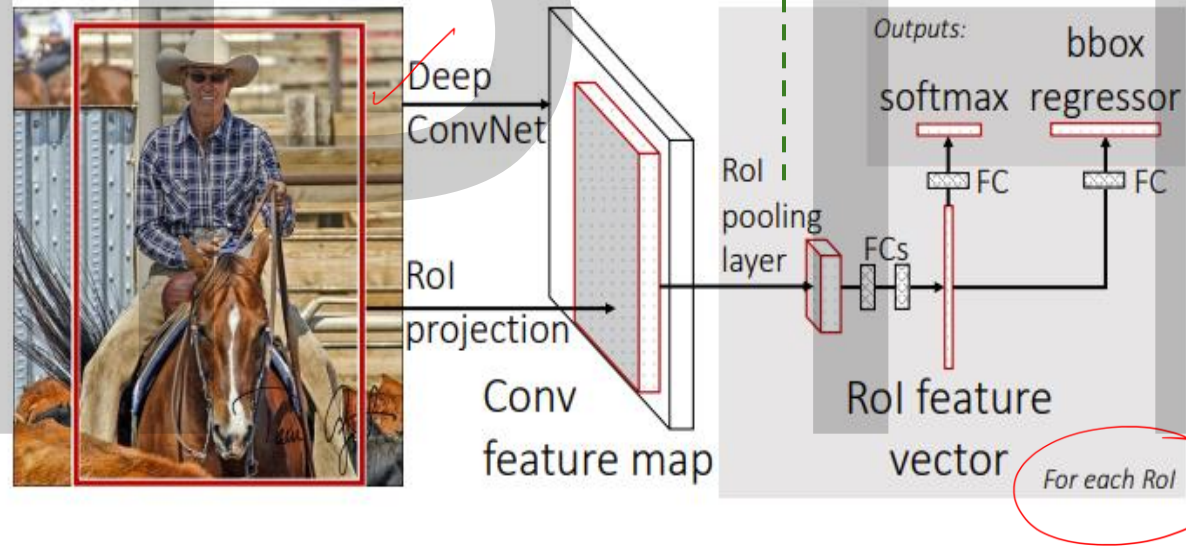
- ✓ **Non-maximum suppression** removes duplicate detections by retaining only regions with the highest scores for each class, based on intersection-over-union (IoU) statistics.
- ✓ It increases accuracy.

Limitations of R-CNN

- ✓ Very slow at Test Time
- ✓ Multi-Stage, Complex Training Pipeline

ROI max pooling takes a region proposal of size $h \times w$ (on the feature map) and converts it to a fixed-size output of $H \times W$ (e.g., 7×7).

Fast R-CNN

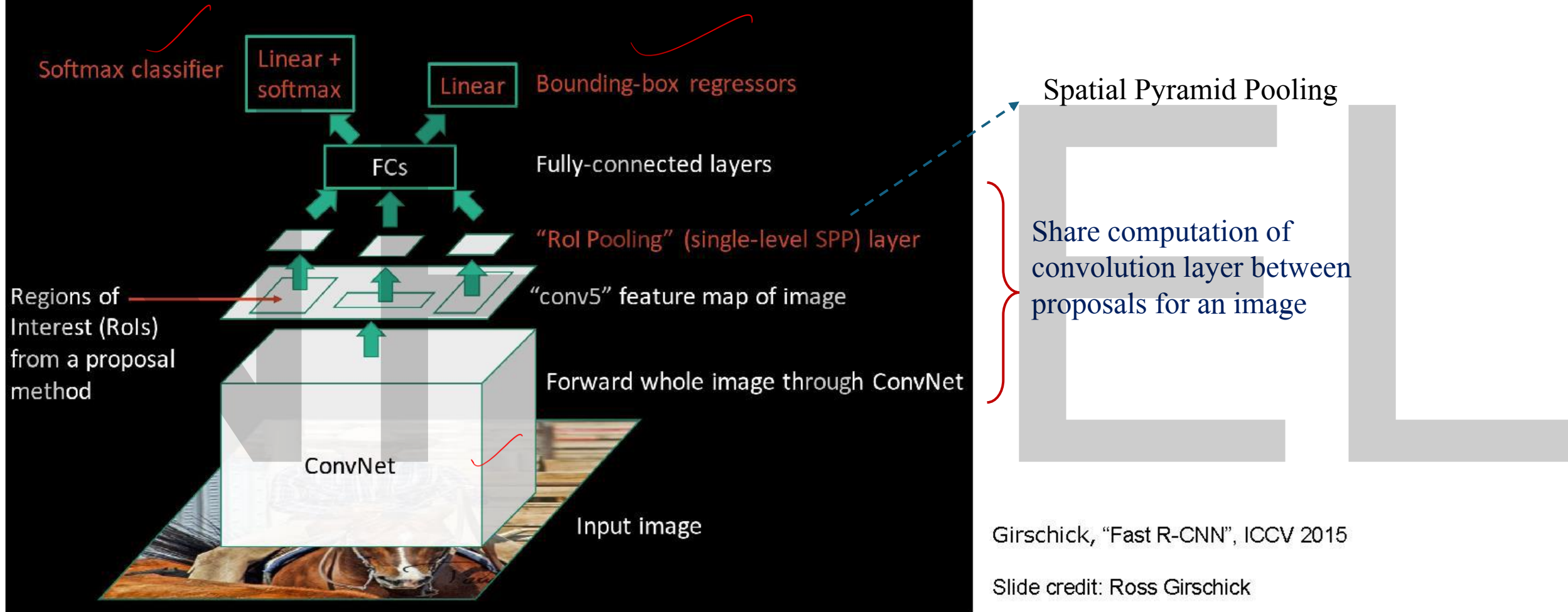


As different ROIs have different shapes and sizes, ROI pooling standardizes them so they can be processed by fully connected layers.

Ref:

Ross Girshick. "Fast R-CNN." In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015, pp. 1440–1448.

Fast R-CNN (test time)



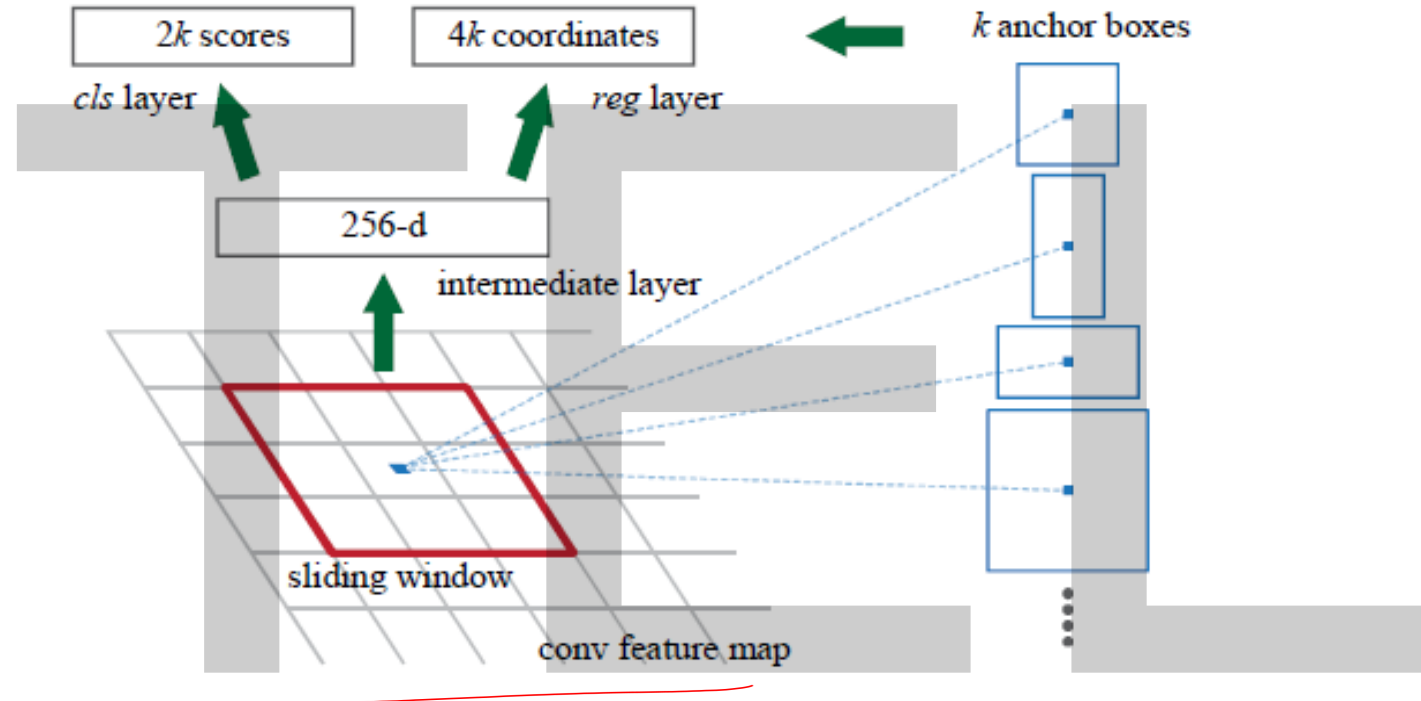
Fast R-CNN made detection much faster than R-CNN by sharing convolutional computations.

Faster R-CNN

It uses **Region Proposal Network (RPN)**



- ✓ Small neural network that **slides over the convolutional feature map** of the entire image.
- ✓ Automatically predicts:
 - Objectness score:*
Where objects could possibly be
 - Bounding box regression offsets:*
The refined bounding box coordinates

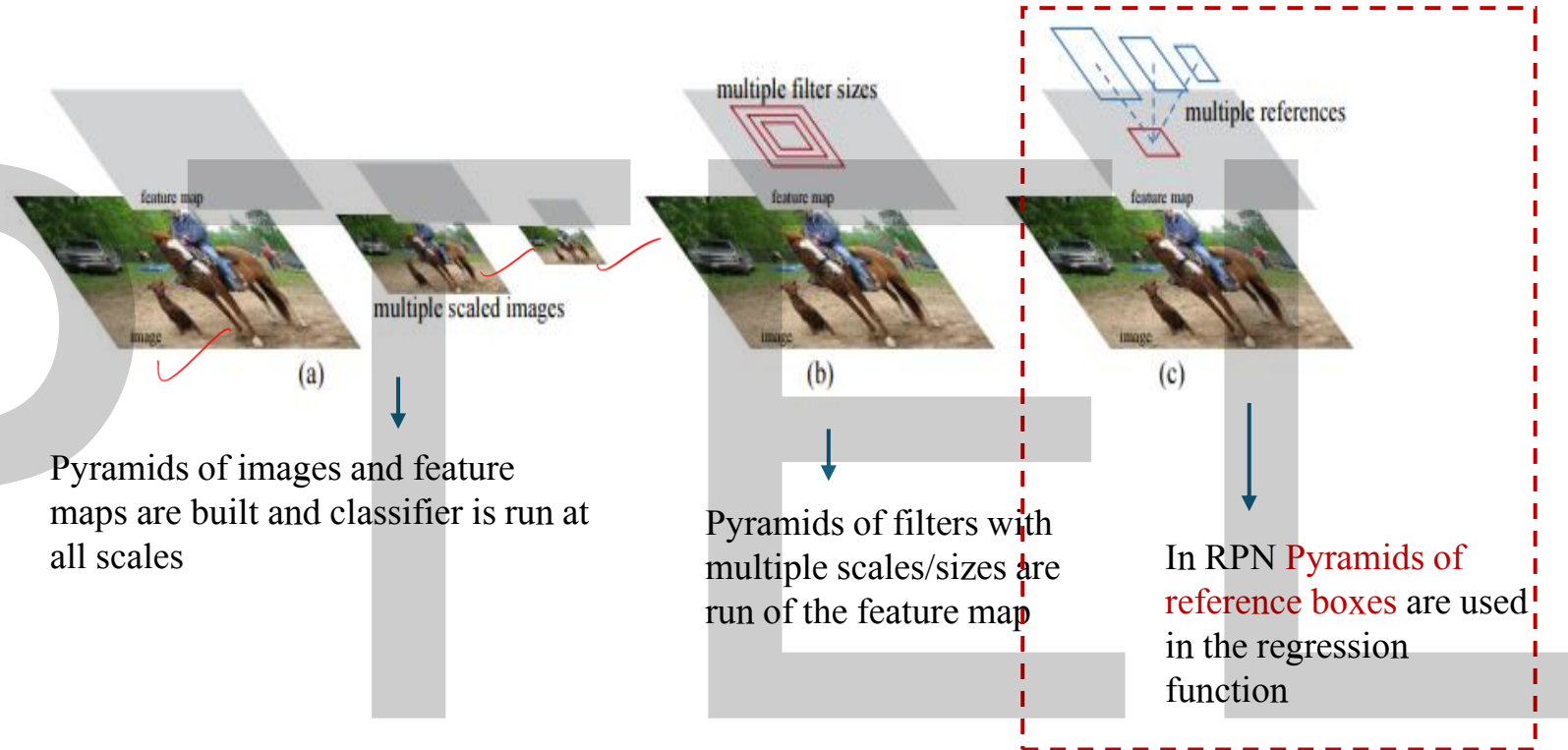
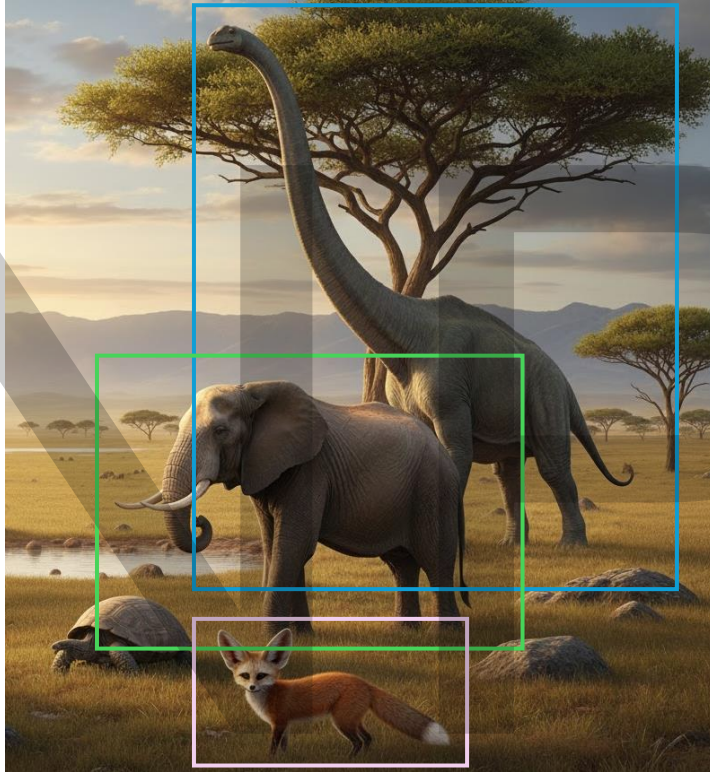


RPN replaces selective search.

Ref:

S. Ren, K. He, R. Girshick and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, (2015)

Objects in images appear in different sizes- How to handle this problem



Ref:

S. Ren, K. He, R. Girshick and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*,

Anchors in Faster R-CNN

A set of predefined bounding boxes (anchors) of

- ✓ multiple scales
- ✓ multiple aspect ratios

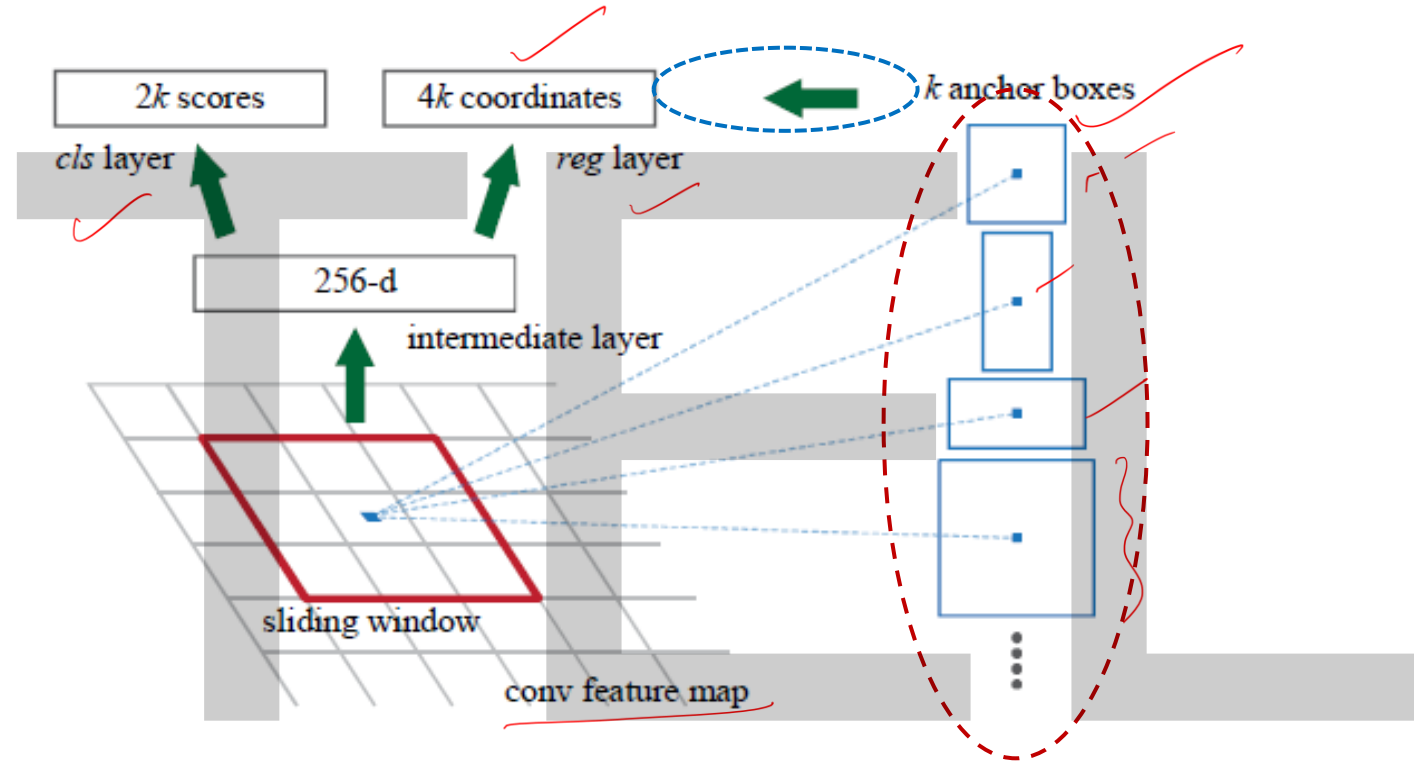
Placed *at every location* of the feature map.
These anchors are called **reference boxes**.

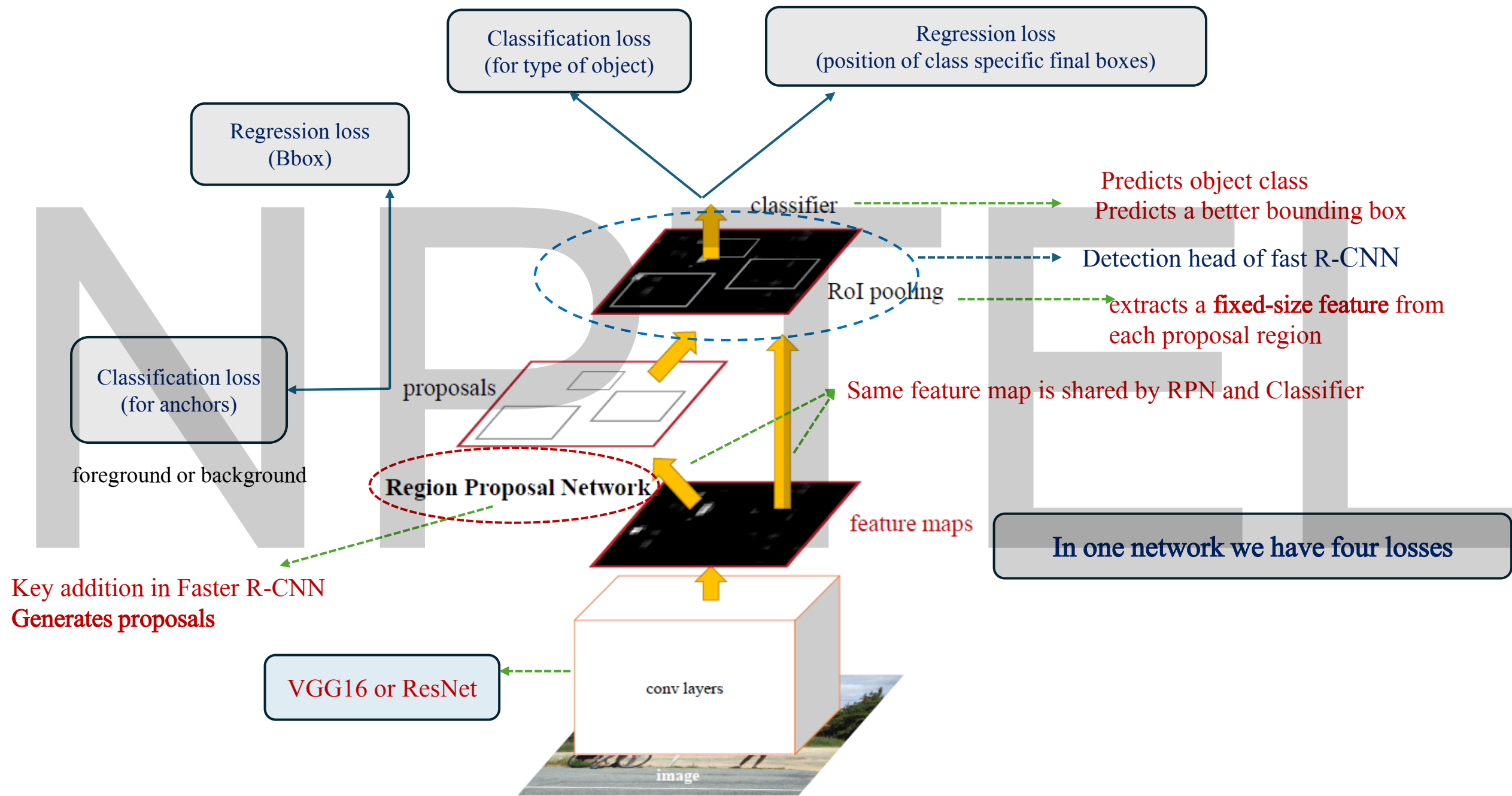
Example (common choice):

3 scales: 128^2 , 256^2 , 512^2

3 aspect ratios: 1:1, 1:2, 2:1

Total 9 anchors per location → **anchor pyramid**.





Performance Comparison

	R-CNN	Fast R-CNN	Faster R-CNN
Testing Time per image (with proposals)	50 secs	2 secs	0.2 secs
mAP (mean Average Precision)	66	66.9	73.2

Faster R-CNN is:

- **Fully trainable end-to-end,**
- **Extremely fast** (region proposals computed in ~10 ms on GPU),
- **More accurate** (learned proposals are better aligned with the detector).

NPTEL

Summary

Next Session

YOLO v1 for Object detection

NPTEL

Thank you